

Series of exercises 3: Sequences of real numbers

Exercise 1 Study the monotonicity and boundedness of the following sequences of real numbers $(u_n)_{n \geq 1}$, where the general term

$$\begin{array}{ll} 1. u_n = \frac{(2n)!}{2^n n!} & 2. u_n = \sum_{k=1}^n \frac{1}{n+k} \\ 3. u_n = \cos \frac{1}{n} & 4. u_n = \ln n - n. \end{array}$$

Exercise 2 Study the convergence of the following sequences of real numbers (u_n) whose general term is given by

$$\begin{array}{ll} 1. u_n = \left(1 + \frac{1}{n}\right)^n ; n \geq 1. & 2. u_n = \ln(n+1) - \ln(n-1); n > 1. \\ 3. u_n = \frac{\sin n}{\sqrt{n}}; n \geq 1. & 4. u_n = \sqrt{n^2 + n} - n; n \geq 0. \end{array}$$

Exercise 3 Given $(u_n)_{n \geq 0}$ a sequence of real numbers whose general term

$$u_n = 2 + \frac{1}{n+1}$$

1. Show that $(u_n)_{n \geq 0}$ is a nondecreasing sequence of real numbers.
2. Show that $(u_n)_{n \geq 0}$ is a bounded sequence of real numbers.
3. Deduce that $(u_n)_{n \geq 0}$ is a convergent and its limit is equal to $\inf_{n \geq 0} (u_n)$.

Exercise 4 Given $(u_n)_{n \geq 1}$ a sequence of real numbers whose general term

$$u_n = \frac{n+1}{2n}.$$

We admit that the sequence has a limit $l = 2$. Find a natural number $n_0 \in \mathbb{N}$ such that all the terms of the sequence u_n having an index $n > n_0$ are in the interval $I =]0.49, 0.51[$.

Exercise 5 Given $(u_n)_{n \geq 0}$ a recursive sequence of real numbers such that $u_0 = 0$ and for all $n \geq 0$

$$u_{n+1} = \frac{2u_n + 1}{u_n + 2}$$

1. By using the mathematical induction, show that $0 \leq u_n < 1$ for all $n \geq 0$.
2. Show that $(u_n)_{n \geq 0}$ is monotonic and convergent.
3. Let $(v_n)_{n \geq 0}$ be a sequence of real numbers such that for all $n \geq 0$

$$v_n = \frac{u_n - 1}{u_n + 1}$$

- (a) Show that (v_n) is a geometric sequence of real numbers. Specify the common ration r and the first term v_0 .
- (b) Express v_n and u_n in function of n and compute $\lim_{n \rightarrow \infty} u_n$.
- (c) Determine the value of natural number N such that for all $n > N$: $u_n > 0.99$.

Exercise 6 Let $(u_n)_{n \geq 1}$ and $(v_n)_{n \geq 1}$ be two sequences of real numbers, such that

$$u_n = \left(\sum_{k=0}^n \frac{1}{k!} \right) + \frac{1}{n \cdot n!} \text{ and } v_n = \sum_{k=0}^n \frac{1}{k!}$$

Show that $(u_n)_{n \geq 1}$ and $(v_n)_{n \geq 1}$ are adjacent sequences.

Exercise 7 Show that the sequence of real number $(u_n)_{n \geq 1}$ defined by $u_{n+1} = k u_n$ where $0 \leq k < 1$ is a Cauchy sequence.

Exercise 8 Extract the convergent subsequences of the sequences of general term for all $n \geq 1$

1. $u_n = (-1)^n \left(1 - \frac{1}{n} \right)$.
2. $u_n = (-1)^n n$.
3. $u_n = 1 - \frac{1}{n}$.
4. $u_n = \sin \left(n \frac{\pi}{2} \right)$.